OWNER	:	**HINDUSTAN PETROLEUM CORPORATION LIMITED (HPCL)**
PMC	:	**ENGINEERS INDIA LIMITED, NEW DELHI (EIL)**
UNIT	:	**504**
PMC JOB NO.	:	**B016**
EQUIPMENT DESCRIPTION	:	**DOCK LEVELER**
EQUIPMENT TAG	:	**504-LZ-930A/B**
P. R. NO.	:	**BO16-RUF-LT-504-MP-PR-0014**
VENDOR NAME	:	**NIHVA TECHNOLOGIES PVT LTD**
VENDOR DOCUMENT NO.	:	**NIHVA_20-21_0157**

DOCUMENT SUBMISSION STATUS – LTHE/VENDOR	**REVIEW STATUS**	
	LTHE	**EIL**
DOC. CATEGORY: (USE TICK MARK) ☑ R ☐ I	☐ Code 1 – No Comments. Proceed With Manufacture / Fabrication / Construction As Per The Document. ☐ Code 2 – Proceed With Manufacture/Fabrication/Construction As Per Commented Document. Revised Document Required. ☐ Review Code 3 – Document Does Not Conform To Basic Requirements As Marked. Resubmit For Review ☐ Code R– Document Is Retained For Records. Proceed With Manufacture / Fabrication ☐ Code V- Void (Submission Required / Not Required)	☐ Code 1 – No Comments. Proceed With Manufacture / Fabrication / Construction As Per The Document. ☐ Code 2 – Proceed With Manufacture/Fabrication/Construction As Per Commented Document. Revised Document Required. ☐ Review Code 3 – Document Does Not Conform To Basic Requirements As Marked. Resubmit For Review ☐ Code R– Document Is Retained For Records. Proceed With Manufacture / Fabrication ☐ Code V- Void (Submission Required / Not Required)
ISSUED FOR ☐ INFORMATION ☑ REVIEW ☐ CONSTRUCTION ☐ AS BUILT	DISCIPLINE / SIGN / DATE	DISCIPLINE / SIGN / DATE

Revision No.	Description	Prepared by	Reviewed by	Approved by	Reviewed by	Approved by	Approved Date
02	Issued for Approval	Shilpa	Amit K	Amit K			23/02/2022
01	Issued for Approval	Shilpa	Amit K	Amit K			24/09/2021

ALL RIGHTS RESERVED

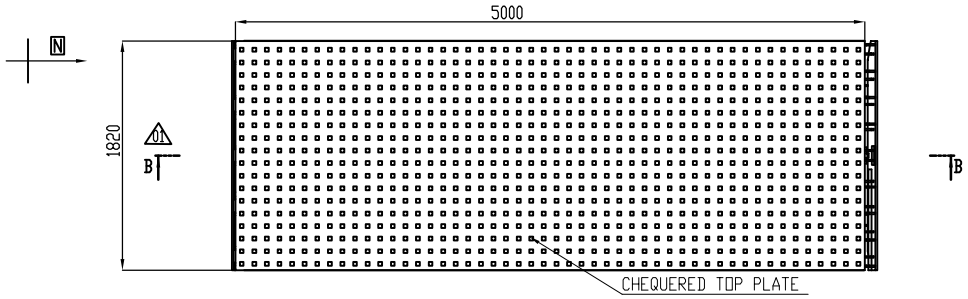
THIS DOCUMENT AND ANY DATA AND INFORMATION CONTAINED THEREIN ARE CONFIDENTIAL AND THE PROPERTY OF L&T HYDROCARBON ENGINEERING (LTHE) AND THE COPYRIGHT THEREIN IS VESTED IN LTHE NO PART OF THIS DOCUMENT, DATA, OR INFORMATION SHALL BE DISCLOSED TO OTHERS OR REPRODUCED IN ANY MANNER OR USED FOR ANY PURPOSE WHATSOEVER, EXCEPT WITH THE PRIOR WRITTEN PERMISSION OF LTHE.

REVISION HISTORY SHEET

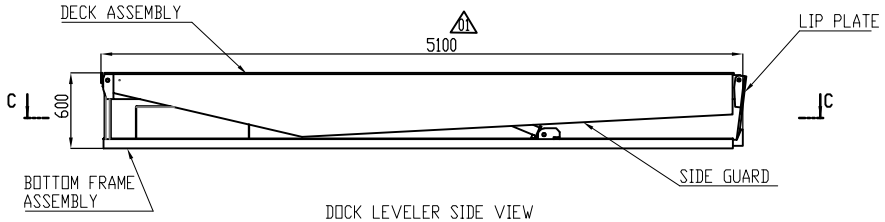
REV. NO.	REV. DATE	REVISED PAGE	REVISION DESCRIPTION
01	24/09/2021	NA	ISSUED FOR APPROVAL
02	23/02/2022	NA	ISSUED FOR APPROVAL

ALL RIGHTS RESERVED

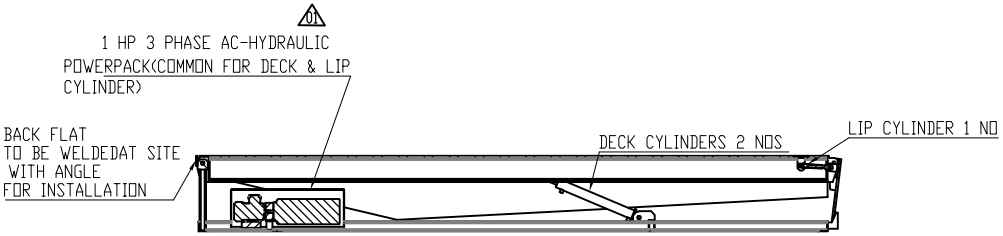
THIS DOCUMENT AND ANY DATA AND INFORMATION CONTAINED THEREIN ARE CONFIDENTIAL AND THE PROPERTY OF L&T HYDROCARBON ENGINEERING (LTHE) AND THE COPYRIGHT THEREIN IS VESTED IN LTHE NO PART OF THIS DOCUMENT, DATA, OR INFORMATION SHALL BE DISCLOSED TO OTHERS OR REPRODUCED IN ANY MANNER OR USED FOR ANY PURPOSE WHATSOEVER, EXCEPT WITH THE PRIOR WRITTEN PERMISSION OF LTHE.



DOCK LEVELER TOP VIEW

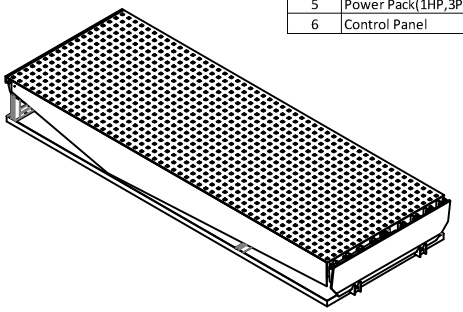


DOCK LEVELER SIDE VIEW

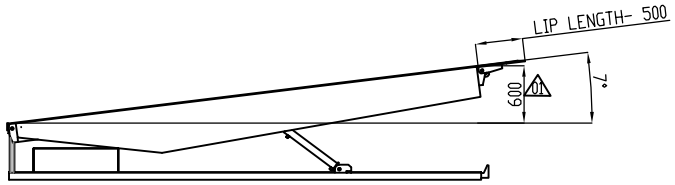


SECTION B-B
SCALE 1 : 15

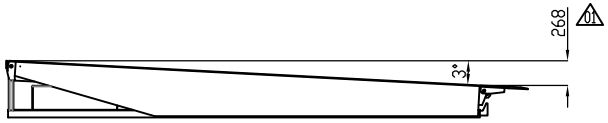
BOM (For Single Dock Leveler)		
SL NO.	DESCRIPTION	QTY
1	Dock Leveler(1820m x 5000mm)	1No
2	Lip (1820 x500mm)	1No
3	Deck Cylinder	2Nos
4	Lip Cylinder	1No
5	Power Pack(1HP,3Phase)	1No
6	Control Panel	1No



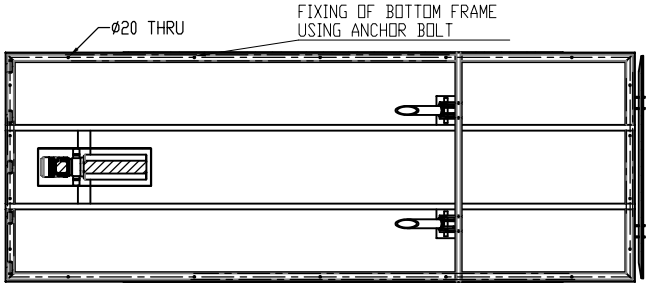
ISOMETRIC VIEW OF DOCK LEVELER



DECK UP HEIGHT - 600 MM



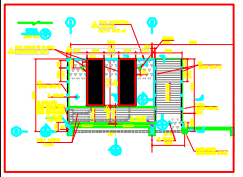
DECK DOWN HEIGHT - 268 MM



SECTION C-C
SCALE 1 : 15

TECHNICAL SPECIFICATIONS OF PIT MOUNTED ELECRO-HYDRAULIC DOCK LEVELER -

1. SIZE OF DOCK LEVELER = 1820mm(W) X 5000mm(L)
2. OVERALL SIZE OF PRODUCT = 1820mm(W)X5100mm(L)X600mm(D)
3. CAPACITY = 6 TON POINT LOAD
4. LIP PLATE SIZE = 500mm(L)X1820mm(W)
5. DECK PLATE = CHEQUERED SHEET 1820mm(W)x5000mm(L)
6. DECK & LIP UP=DOWN OPERATION BY HYDRAULIC CYLINDER
7. DECK UP HEIGHT= (+)600 MM
8. DECK DOWN HEIGHT= (-)268MM 02
9. HEIGHT OF DOCK LEVELER FROM ROAD LEVEL= 900 mm,
10. OPERATION CONTROL BY PUSH BUTTONS ON CONTROL PANEL MOUNTED ON STAND NEAR DOCK LEVELER
11. ELECTRICAL SUPPLY OF EQUIPMENT= 415 VOLT 3 PHASE AC , 50HZ
12. PAINT = FIROZI BLUE
13. MATERIAL OF CONSTRUCTION = MILD STEEL AS PER IS STANDARD
14. MANUFACTURING STANDARD OF DOCK LEVELER =EN1398
15. ELECRICAL MOTOR COMPLIANCE ZONE 1-IIC-T3 HAZARDOUS AREA
16. PIT SIZE = 1870 mm(W)X5100mm(L)x600 mm(D)
17. DOCK LEVELER FIXING IN PIT BY ANCHOR BOLT .
18. REQUIRED PIT & ALL CIVIL WORK FOR MOUNTING OD DOCK LEVELER IS IN LTHE SCOPE .
    REFER PIT DRAWING FOR CONSTRUCTION OF PIT B016-RUF-LT-504-MP-DW-NTPL(1)-13005
19. HYDRAULIC CYLINDERS & POWERPACK IS DESIGNED FOR LIFTING BARE DOCK ONLY.
20. MOTOR RATING=1HP 3 PHASE AC-HYDRAULIC
21. All ELECTRICAL EQUIPMENTS SHALL HAVE PAINT SHADE RAL 7031 02
22. ALL FLAME PROOF CONTROL PANEL 360X360X210MM 02
23. WEIGHT= APPROX 2500KG EMPTTY & 8500KG OPERATING 02
24. WE WILL PROVIDE FRP CANOPY FOR CONTROL PANEL ONLY. 02



504-LZ-930A/B	CIVIL DRAWING OF DOCK LEVELER
DRAWING NO.	DRAWING TITLE
REFERENCE DOCUMENTS / DRAWINGS	
ISSUED FOR: AS BUILT / CONSTRUCTION / INFORMATION / REVIEW	
01	ISSUED FOR APPROVAL
REV	DESCRIPTION
OWNER	HINDUSTAN PETROLEUM CORPORATION LIMITED VISAKH REFINERY MODERNIZATION PROJECT EPCC-03 PACKAGE FOR RESIDUE UP-GRADATION FACILITY (RUF)
PMC	EngineersIndia Limited
LSTK CONTRACTOR	L&T Hydrocarbon Engineering
DETAILED ENGINEERING CENTER	L&T - CHIYODA LIMITED
VENDOR	NIRVA TECHNOLOGIES PVT LTD
VENDOR PO NO	REFINARY/75000-82256/VNM
VENDOR DWG. NO	NBHVA_20-21_0157
DATE 31.05.21	SCALE 1:1
TITLE	GAD FOR DOCK LEVELER (504-LZ-930A/B)
DEPT. DESIGN	
DRN BY: SNS	PROJECT: L&T-RUF
CHK BY: VJ	PRO NO.:
APP BY: AK	DRG. NO.:
SH.1 OF 1	
REV. 02	

L&T - CHIYODA LIMITED

DATA SHEET FOR DOCK LEVELER (504-LZ-930A/B)

Sr No	Parameter	Remark
1	Required Capacity	06 Ton
2	Leveler operation mode	AC power pack hydraulically operated
3	Lip operation mode	hydraulic
4	Operation control	by push buttons and control panel
5	Height of dock leveler from road Level	900 mm
6	Travel above & below dock level	600 mm & 268 mm
7	Platform Size (W x L)	1820 mm x 5000 mm
8	Lip size (W x L)	1820 mm x 500 mm
9	Maximum slope can be achieved	7 Degree
10	Minimum slope can be achieved	3 Degree

* Please refer Attachment-1 for standard catalog of Dock Leveler.

ALL RIGHTS RESERVED

THIS DOCUMENT AND ANY DATA AND INFORMATION CONTAINED THEREIN ARE CONFIDENTIAL AND THE PROPERTY OF L&T HYDROCARBON ENGINEERING (LTHE) AND THE COPYRIGHT THEREIN IS VESTED IN LTHE NO PART OF THIS DOCUMENT, DATA, OR INFORMATION SHALL BE DISCLOSED TO OTHERS OR REPRODUCED IN ANY MANNER OR USED FOR ANY PURPOSE WHATSOEVER, EXCEPT WITH THE PRIOR WRITTEN PERMISSION OF LTHE.